

HDOC – Day 8 Activity

Question 1:

Write a C program to find the surface area and volume of a sphere in square millimeters and cubic millimeters when radius is given in centimeters.

- **Step 1: Understanding the Problem**

- **Required Input:**

Radius of the sphere in centimeter(cm).

Output :

- Surface Area in square millimeter (mm^2)
- Volume in cubic millimeters (mm^3)

Formula:

Convert radius from cm to mm:

Radius in mm = Radius in cm $\times$ 10

Surface area of sphere:

Surface Area = $4 \times \pi \times r^2$

Volume of sphere:

$$\text{Volume} = (4/3) \times \pi \times r^3$$

where r is the radius in millimeters.

Take:

$$\pi = 3.14$$

Test Case	Input Radius	Expected Surface Area	Expected Volume	Actual Output	Result
1	1	1256.64	4188.79	1256.64,4188.79	Pass
2	2	5026.54	33510.29	5026.54,33510.29	Pass
3	3	31415.90	523598.33	31415.90,523598.33	Pass

- C Programming:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float radius_cm, radius_mm;
```

```
    float surface_area, volume;
```

```
    const float PI = 3.14159;
```

```
    printf("Enter radius in centimeters: ");
```

```
    scanf("%f", &radius_cm);
```

```
radius_mm = radius_cm * 10;

surface_area = 4 * PI * radius_mm *
radius_mm;
    volume = (4.0 / 3.0) * PI * radius_mm *
radius_mm * radius_mm;
    printf("Surface Area = %.2f square
millimeters\n", surface_area);
    printf("Volume = %.2f cubic
millimeters\n", volume);

    return 0;
}
```

- **Step 2: Algorithm**

1. Start
2. Declare variables radius_cm, radius_mm, surface_area, and volume.
3. Read the radius in centimeters.

4. Convert the radius from centimeters to millimeters using:
 $\text{radius_mm} = \text{radius_cm} \times 10.$
5. Calculate surface area using:
 $\text{surface_area} = 4 \times \pi \times \text{radius_mm} \times \text{radius_mm}.$
6. Calculate volume using:
 $\text{volume} = (4/3) \times \pi \times \text{radius_mm} \times \text{radius_mm} \times \text{radius_mm}.$
7. Display the surface area in mm^2 .
8. Display the volume in mm^3 .
9. Stop

Compilation and Execution:

Test Case 1:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter radius in centimeters:1  
Surface Area = 1256.00 square millimeters  
Volume = 4186.67 cubic millimeters
```

Test Case 2:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter radius in centimeters:2  
Surface Area = 5024.00 square millimeters  
Volume = 33493.34 cubic millimeters
```

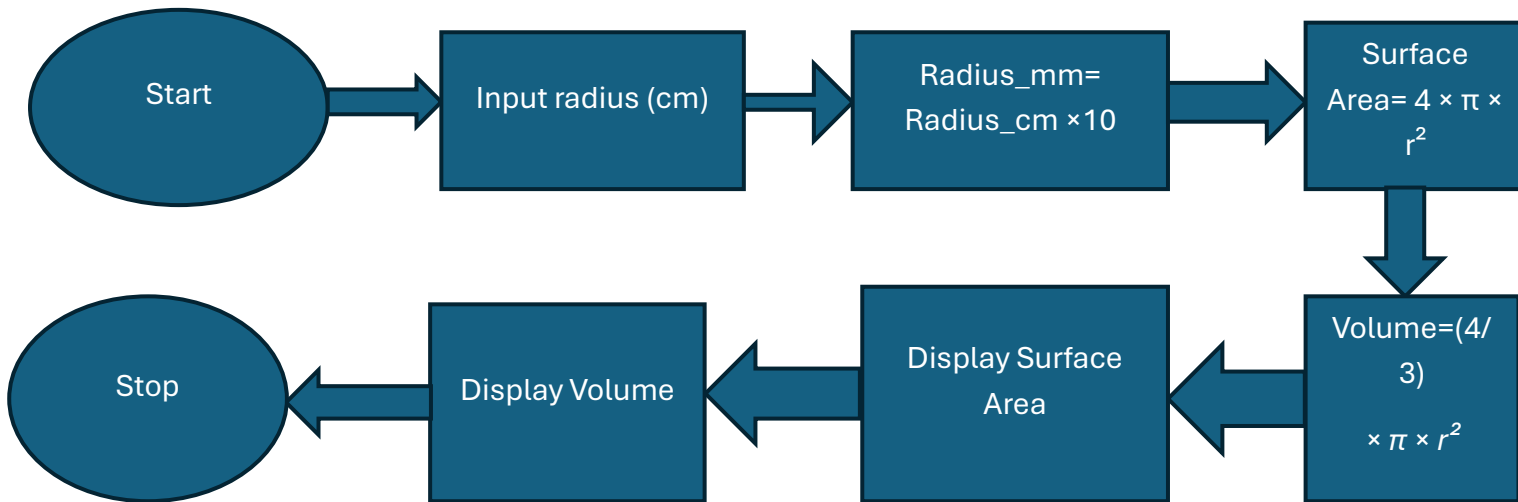
Test Case 3:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter radius in centimeters:5  
Surface Area = 31400.00 square millimeters  
Volume = 523333.34 cubic millimeters
```

Compilation and Execution Screenshots:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter radius in centimeters:1  
Surface Area = 1256.00 square millimeters  
Volume = 4186.67 cubic millimeters  
  
(kali㉿kali)-[~]  
$ ./a.out  
Enter radius in centimeters:2  
Surface Area = 5024.00 square millimeters  
Volume = 33493.34 cubic millimeters  
  
(kali㉿kali)-[~]  
$ ./a.out  
Enter radius in centimeters:5  
Surface Area = 31400.00 square millimeters  
Volume = 523333.34 cubic millimeters
```

Step 3: Flowchart:



Flowchart Evaluation:

For Every test case, the flowchart follows the same sequence.

Therefore, the flowchart correctly represents the algorithm.

QUESTION 2:

Write a C program to find the area, arc length, and perimeter of a quadrant when radius is given in meters and output is required in centimeters.

Step 1: Understanding the Problem

Required Input:

Radius of the quadrant in meters (m).

Required Output:

Area in square centimeters (cm^2)

Arc length in centimeters (cm)

Perimeter in centimeters (cm)

Conversion

Since:

1 meter = 100 centimeters

Therefore:

Radius in cm = Radius in m $\times$ 100

Formula

For a quadrant:

Area = $\pi r^2 / 4$

Arc Length = $\pi r / 2$

Perimeter = Arc Length + 2r

where r is the radius in centimeters.

Take:

$\pi = 3.14$

Step 2:

Test Case Table:

Test Case	Input Radius(cm)	Expected Area(cm ²)	Expected Arc Length(cm)	Expected Perimeter(cm)	Result
1	1	7853.98	157.08	357.08	Pass
2	2	31415.90	314.16	714.16	Pass
3	5	196349.38	785.40	1785.40	Pass

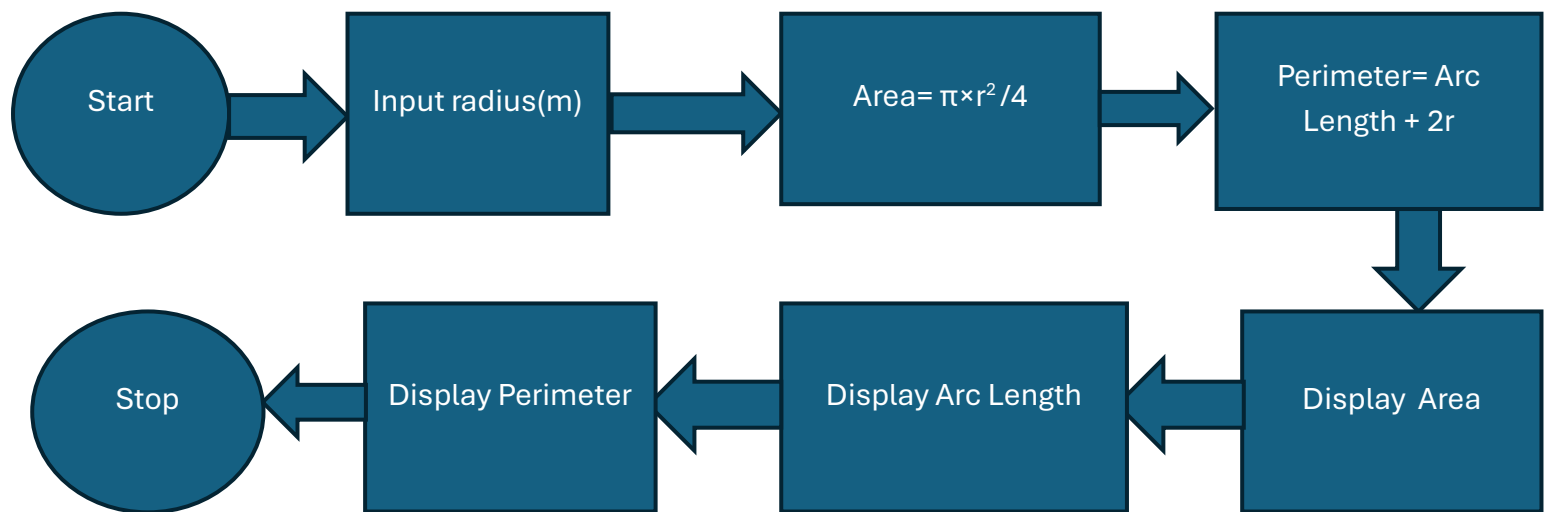
Step 3: Algorithm

- Start
- Declare variables radius_m, radius_cm, area, arc_length, and perimeter.
- Read the radius in meters.
- Convert radius from meters to centimeters:

`radius_cm = radius_m  $\times$  100.`

- Calculate the area:
 $\text{area} = \pi \times \text{radius_cm} \times \text{radius_cm} / 4.$
- Calculate the arc length:
 $\text{arc_length} = \pi \times \text{radius_cm} / 2.$
- Calculate the perimeter:
 $\text{perimeter} = \text{arc_length} + 2 \times \text{radius_cm}.$
- Display area in cm^2 .
- Display arc length in cm.
- Display perimeter in cm.
- Stop

• Step 4: Flowchart



• C Programming:

```

#include <stdio.h>

int main()
{
    float radius_m, radius_cm;

    float area, arc_length, perimeter;

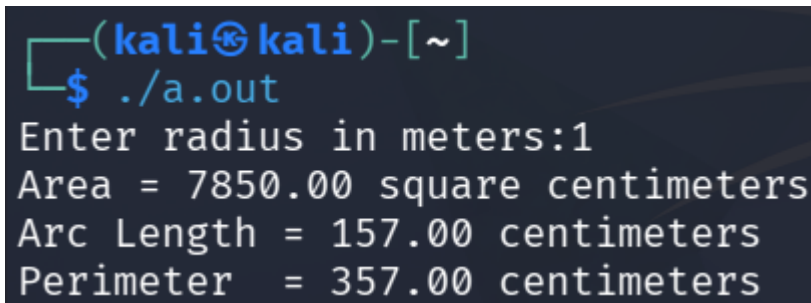
    const float PI = 3.14159;
  
```



```
printf("Enter radius in meters: ");  
scanf("%f", &radius_m);  
  
radius_cm = radius_m * 100;  
  
area = (PI * radius_cm * radius_cm) / 4;  
arc_length = (PI * radius_cm) / 2;  
perimeter = arc_length + (2 * radius_cm);  
  
printf("Area = %.2f square centimeters\n", area);  
printf("Arc Length = %.2f centimeters\n", arc_length);  
printf("Perimeter = %.2f centimeters\n", perimeter);  
  
return 0;  
}
```

Compilation and Execution:

Test Case 1:



```
(kali㉿kali)-[~]  
$ ./a.out  
Enter radius in meters:1  
Area = 7850.00 square centimeters  
Arc Length = 157.00 centimeters  
Perimeter = 357.00 centimeters
```

Test Case 2:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter radius in meters:2  
Area = 31400.00 square centimeters  
Arc Length = 314.00 centimeters  
Perimeter  = 714.00 centimeters
```

Test Case 3:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter radius in meters:5  
Area = 196250.00 square centimeters  
Arc Length = 785.00 centimeters  
Perimeter  = 1785.00 centimeters
```

Compilation and Execution Screenshot:

```
(kali㉿kali)-[~]
```

```
$ ./a.out
```

Enter radius in meters:1

Area = 7850.00 square centimeters

Arc Length = 157.00 centimeters

Perimeter = 357.00 centimeters

```
(kali㉿kali)-[~]
```

```
$ ./a.out
```

Enter radius in meters:2

Area = 31400.00 square centimeters

Arc Length = 314.00 centimeters

Perimeter = 714.00 centimeters

```
(kali㉿kali)-[~]
```

```
$ ./a.out
```

Enter radius in meters:5

Area = 196250.00 square centimeters

Arc Length = 785.00 centimeters

Perimeter = 1785.00 centimeters